

Sudharshan Ram Jayaraman

Data & AI Intern Engineer

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PROFILE

Master's student in Computational Science and Engineering at TU Munich with practical experience in **Data Engineering** and **Machine Learning**. Gained hands-on experience at Hilti creating **data pipelines** with **Microsoft Fabric** and **Python**. Skilled in building AI agents, cleaning sensor data, and developing predictive models. Seeking an internship to apply expertise in Data Analytics, Automation, or AI.

EDUCATION

Technical University of Munich (TUM) <i>Master of Science in Computational Science and Engineering</i>	Munich, Germany <i>Oct 2024 – Present</i>
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- **Focus:** Deep Learning, Machine Learning, Parallel Programming, Visual Data Analytics.
- **Current Status:** Active student specializing in data-driven analysis and numerical modeling.

Amrita Vishwa Vidyapeetham <i>Bachelor of Technology in Mechanical Engineering</i>	India <i>2018 – 2022</i>
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- **CGPA:** 9.3/10 (Excellent Academic Standing).
- **Relevant Coursework:** Finite Element Methods (FEM), CFD, Thermodynamics

TECHNICAL SKILLS

Languages:	Python (Advanced), C++, SQL, MATLAB, LaTeX.
Data & AI:	PyTorch, NumPy, Pandas, Scikit-learn, Spark, Hadoop.
Data Engineering:	Microsoft Fabric, ETL Pipelines, Data Lakehouse, SQL, Dataiku, n8n.
Tools & Cloud:	Git/GitHub, Docker, JIRA, Confluence, SharePoint, Tableau, Power BI.
Engineering:	ANSYS Workbench, Fluent, Catia, Fusion 360.

WORK EXPERIENCE

Hilti Entwicklungsgesellschaft GmbH <i>Working Student - Data Analyst</i>	Germany <i>May 2025 – Present</i>
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- **Data Pipeline Engineering:** Designed and implemented end-to-end data pipelines using **Microsoft Fabric**, consolidating NFC sensor data and CSV logs from SharePoint into a Data Lakehouse.
- **Data Processing:** Performed data cleaning, feature engineering, and preprocessing using **Python (Pandas, NumPy)** to create curated datasets for downstream analytics.
- **Machine Learning:** Applied **DBSCAN clustering** for anomaly detection in battery cell voltages data and developed predictive models using **Scikit-learn**.
- **Automation:** Streamlined reporting with automated workflows and supported Kanban model automation via JIRA. The streamlined data pipeline improved **process efficiency by 62%**. Enabled future fault mitigation in **cell voltage anomalies** using the identified clusters.
- **Soft Skills:** Collaborated with cross-functional teams to define data requirements while working independently on tasks like preprocessing, evaluating insights and optimizing workflows.

Tata Technologies <i>Design Engineer</i>	India <i>Jun 2022 – Oct 2023</i>
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- **Engineering Design & PLM:** Designed chassis components(brackets, cables, holding mount, calipers) and kinematic modeling of steering mechanisms using **Catia V5**; managed design instances and product lifecycles via **Teamcenter** for Volvo models (XC90, EX30).
- **Analysis:** Reverse engineered 3D models from raw scan data for kinematic analysis; this project had a positive feedback from the client (Volvo).

- **Optimization & Impact:** Owned the Pad Wear Indicator (PWI) package, routing brake cable and plastic modeling (clip assembly); design optimizations contributed to an **18% profit increase** for the project.
- **Soft Skills:** Demonstrated **Agile leadership** and **stakeholder management** by leading team of design engineers and delivering project milestones under tight deadlines.

KEY PROJECTS

Multi-Agent E-commerce Automation

(AI, LLM, Python)

- **Agent Orchestration:** Architected a multi-agent AI solution to manage customer queries and product recommendations, building pipeline logic to coordinate agents for higher response relevance.
- **System Integration:** Integrated data retrieval systems and context management to enable personalized assistance across the e-commerce lifecycle.
- **Learning Outcome:** Design of agentic workflows and LLM application development.

Machine Learning in Crowd Modelling

(Python, PyTorch, Vadere, Git)

- **Hybrid Modeling:** Integrated Machine Learning with dynamical systems (SIR epidemiology dynamics) to simulate and analyze complex crowd behavior.
- **Advanced Analytics:** Applied dimensionality reduction (PCA, Diffusion Maps) and latent-space modeling (VAEs) to reconstruct data-driven dynamical systems from motion data.
- **Learning Outcome:** Machine learning application in physics-based simulations. Dimensionality reduction techniques for large datasets.

Parallel CFD Solver Development

(C++, MPI, GPU, OpenMP, Git)

- **HPC Development:** Developed a 2D Navier-Stokes solver from scratch, implementing numerical schemes on a staggered grid using finite differences.
- **Optimization:** Achieved high-performance parallelization using **MPI** with domain decomposition and **GPU acceleration**; extended the solver to Multigrid methods (V-cycle, W-cycle).
- **Learning Outcome:** Parallel computing patterns, memory management in C++, and software development lifecycles.

Visual Data Analysis of US Sales

(Tableau, Data Analytics)

- **Dashboarding:** Designed interactive Tableau dashboards to visualize US sales trends, implementing dynamic filters for region and time periods.
- **Statistical Analysis:** Conducted clustering analysis and trend line evaluation to study the effect of visual channels on human perception.
- **Learning Outcome:** Business Intelligence (BI) storytelling and user-centric dashboard design.

CERTIFICATIONS

- **Data Engineering:** Spark, Hadoop Snowflake (Coursera); Databricks Fundamentals.
- **Machine Learning:** Applied Machine Learning in Python (Coursera); 100 Days of Code (Python).
- **Generative AI:** Prompt Engineering Data Analysis with LLM.
- **Database:** SQL Intermediate Certification (HackerRank).

LANGUAGES

English: C1 (Professional Working Proficiency)

German: B2 (Independent User)

Tamil: Native